Lab 5 — Steady State and Eigen

Friday, November 20 (~20 min)

Last updated: October 29, 2026.

In class Friday, November 20 (same day as Lecture 30 / Quiz 10). Quiz 10 is Week 11 inverses / rank / determinants — **not** this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

Payoff. Lab 4 asked why different starts share a long-run mix. Today: that mix is the $\lambda = 1$ **eigenvector** (scaled); the other eigenvalue has $|\lambda| < 1$, so its mode fades. Same story as Lecture 30's dominant-eigenbehavior fill-in.

Learning objectives

1. Find a steady state of $x = Ax$ (or an eigenvector for $\lambda = 1$).
2. Use software eigenvalues/vectors and match them to the Lab 4 long-run mix.
3. State how an eigen-direction explains repeated multiplication by A ($A^t v = \lambda^t v$).

Math connections

Eigenvalues and eigenvectors (Lectures 35–37); build on Lab 4's iteration.

Key syntax (this lab)

Task | Octave | Python |